

Automated BI Report Delivery to WhatsApp

Detailed limitations, risks, and implementation controls for Looker, Metabase, browser automation, and WhatsApp messaging

Core conclusion

The project is feasible when reports can be rendered through supported BI features and delivered through an official WhatsApp route. The largest likely blocker is a requirement to post into an existing ordinary WhatsApp group. Correctness, data privacy, and reliable operations are more important than simply taking a screenshot.

Prepared for project research, feasibility review, technical planning, and management discussion

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Table of contents

1. Executive summary	3
2. What the assignment is asking for	3
2.1 The six jobs hidden inside one sentence	3
3. Recommended system design	4
3.1 Recommended source order	4
3.2 Recommended WhatsApp order	4
4. Top risk register	5
5. Detailed limitations and controls	6
A. Requirements and feasibility	6
B. Report source, authentication, and networking	6
C. Rendering and data correctness	8
D. Browser automation limitations	10
E. WhatsApp platform and messaging limitations	11
F. Security, privacy, and governance	13
G. Scheduling and reliability	14
H. User experience and communication	16
I. Operations, maintenance, and cost	17
6. Platform-specific implementation approach	19
6.1 Looker	19
6.2 Metabase	19
6.3 Playwright fallback	19
6.4 WhatsApp delivery	19
7. Failure scenarios and expected response	20
8. Phased implementation plan	21
9. Testing strategy	22
10. Production acceptance criteria	22
11. Questions to ask the manager and platform owners	22
12. Final recommendation	23
13. Glossary	24
14. References	25

1. Executive summary

Go/no-go decision

GO when the project can use official BI exports and official one-to-one WhatsApp delivery to approved recipients, or when the organization is confirmed eligible for an API-managed group. CONDITIONAL when exact browser screenshots are required. NO-GO for production when the only acceptable destination is an existing ordinary group and only unsupported web automation can reach it.

The requested bot sounds simple: open a dashboard, take a screenshot, and send it to WhatsApp. In practice, it is a chain of systems with different security and reliability rules. A failure in any link can produce a missing report, a duplicate report, an incorrect report, or a data leak.

The safest approach is to build a reporting pipeline rather than one long automation script. Each stage should have a clear input, output, validation rule, timeout, status, and owner. The service should prefer native Looker or Metabase delivery capabilities, use browser automation only for gaps, and use an official WhatsApp Business route for production.

The most dangerous failure is not a visible crash. It is a normal-looking screenshot that contains stale data, the wrong filter, the wrong customer, or an incomplete chart. For this reason, the bot must fail closed: if freshness and completeness cannot be proven, it must not send the report.

- Resolve the WhatsApp destination model before development.
- Prove one real report export and one production-like WhatsApp media delivery early.
- Use a dedicated least-privilege automation identity, not a normal employee login.
- Validate data freshness, report identity, filters, chart completion, media size, and recipient authorization.
- Use durable scheduling, idempotency, per-recipient delivery status, alerts, and short artifact retention.
- Run a controlled pilot before sending confidential production reports.

2. What the assignment is asking for

The assignment is to replace a repetitive human task. Today, someone may log into a BI tool before a stand-up meeting, wait for charts to load, take a screenshot, and paste it into WhatsApp. The proposed bot performs those steps automatically on a schedule.

2.1 The six jobs hidden inside one sentence

- Trigger the work at the correct business time or when data is ready.
- Authenticate to the reporting platform and obtain the correct report.
- Render the report as a PNG or PDF without missing charts.
- Validate that the content is complete, fresh, correctly filtered, and safe to distribute.
- Deliver the media through a supported WhatsApp mechanism and record the result.
- Monitor failures, prevent duplicates, clean up files, and provide a manual recovery process.

Important distinction

A bot that can click and send is only a prototype. A production service must also prove correctness, authorization, delivery, and recovery.

3. Recommended system design

Recommended reporting pipeline



Every stage records a job ID so failures can be retried without sending duplicates.

Use one source adapter per BI platform and one delivery adapter per messaging provider. The core pipeline should not know whether the report came from Looker, a Metabase alert, a mailbox attachment, or Playwright. This separation makes provider changes and testing manageable.

3.1 Recommended source order

Source	Preferred path	Why
Looker dashboard	Official Render API or scheduled PNG/PDF delivery	Supported asynchronous render completion and stable dashboard IDs [1][2].
Metabase single chart	Scheduled alert webhook	Webhook includes visualization as base64 PNG plus raw data [4].
Metabase full dashboard	Scheduled PDF subscription to bot mailbox/Slack	Native whole-dashboard delivery; webhook is not currently available for subscriptions [3][4].
Exact on-screen PNG	Playwright as controlled fallback	Needed only where native output cannot meet the approved visual requirement.

3.2 Recommended WhatsApp order

- First choice: official Cloud API or Twilio, sending the same approved media template separately to every opted-in recipient.
- Second choice: official Groups API only after eligibility and group-size requirements are confirmed in the real account [7].
- Fallback: email or Slack when official WhatsApp delivery cannot meet governance or group requirements.
- Prototype only: WhatsApp Web emulation with a separate test number and no sensitive production data.

4. Top risk register

The following risks deserve management attention before implementation. Detailed explanations and controls appear in Section 5.

#	Risk	Level	Required decision/control
1	Existing group is not eligible for official automation	CRITICAL	Decide group vs individual delivery before coding
2	Sensitive screenshots leave BI access controls	CRITICAL	Data-owner/privacy approval and minimal content
3	Stale or partial data is delivered	CRITICAL	Freshness and tile-completion validation
4	SSO/MFA blocks the automation account	CRITICAL	Approved service authentication
5	Scheduled WhatsApp message lacks opt-in/template	CRITICAL	Consent records and approved media template
6	Wrong filter/timezone/user context	CRITICAL	Explicit config and visible assertions
7	Clock fires before ETL is complete	CRITICAL	Data-ready event or freshness gate
8	Retries or replicas send duplicates	HIGH	Idempotency key and durable job ledger
9	Dashboard layout changes break capture	HIGH	Native API or stable capture contract
10	Recipient list is outdated	HIGH	Owned access list and periodic review
11	Image is rejected or unreadable	HIGH	Validate, split, compress, phone-test
12	No owner responds to failures	CRITICAL	Named business and technical owners

5. Detailed limitations and controls

Each limitation is explained in simple terms using four questions: what it means, what can go wrong, how to reduce the risk, and what risk still remains. Severity reflects the likely effect on feasibility, correctness, security, or operations.

A. Requirements and feasibility

1. The phrase 'post screenshots on WhatsApp' is not specific enough

CRITICAL

What it means: The project changes completely depending on whether the destination is one person, many individual recipients, a new API-managed group, or an existing team group. It also matters whether the output is one chart, one dashboard tab, a full dashboard, a PNG, or a PDF.

What can go wrong: The team may build a technically good bot that cannot send to the required destination, captures the wrong report, or produces an unreadable format.

How to reduce the risk: Write a one-page requirements contract before coding. Name the exact dashboards, tabs, filters, schedule, timezone, recipients, output format, acceptable delay, data classification, and fallback channel. Include sample screenshots approved by the report owner.

Risk that remains: Business needs can still change, so configuration and ownership must be explicit rather than embedded in code.

2. An existing ordinary WhatsApp group may be impossible to automate officially

CRITICAL

What it means: The official Groups API now exists, but current access is restricted. Provider documentation states that the business must have an Official Business Account and at least the 100,000 messaging limit, and each group is capped at eight participants [7]. This is not the same as controlling any existing group created in the consumer app.

What can go wrong: If the assignment specifically requires posting into a normal large team group, the production design may be blocked before development begins.

How to reduce the risk: Check the company's WhatsApp Business Account eligibility in the actual Meta environment. If it does not qualify, choose individual official messages, Slack/email, or obtain written acceptance for a non-production web-emulation prototype.

Risk that remains: Eligibility and platform rules can change. The delivery adapter must remain replaceable.

3. A screenshot may not be the right report format

MEDIUM

What it means: A screenshot preserves the visual layout, but it may crop tables, hide scrollable rows, compress small labels, and make numbers inaccessible to screen readers. A PDF or a concise message plus link may be more useful.

What can go wrong: Recipients receive something that technically meets the request but is difficult to read or interpret on a phone.

How to reduce the risk: Ask users to compare a phone-sized PNG, a multi-page PDF, and a short KPI summary. Choose the smallest format that preserves the information needed for the meeting.

Risk that remains: Complex dashboards may need multiple tailored outputs rather than one universal image.

B. Report source, authentication, and networking

4. Single sign-on, MFA, CAPTCHA, and conditional access block unattended login

CRITICAL

What it means: Enterprise dashboards commonly redirect through Okta, Microsoft Entra, Google Workspace, or another identity provider. The login may require a phone prompt, security key, CAPTCHA, device compliance, or a trusted location. A scheduled service cannot ask a person to approve every login.

What can go wrong: The bot works during a demo but stops when the session expires or a security policy changes.

How to reduce the risk: Prefer official API credentials or a dedicated automation service account with least privilege and an approved non-interactive authentication flow. Never disable MFA for a normal employee account merely to make the bot work. Escalate the design to identity and security owners.

Risk that remains: Even service credentials expire or are rotated; monitoring and a renewal runbook remain necessary.

5. VPN, private DNS, firewall, or IP allowlists can isolate the dashboard

HIGH

What it means: The report URL may only be reachable inside the company network. A container running in a public cloud may not resolve the hostname or may be rejected because its outbound IP is not approved.

What can go wrong: Development succeeds on an employee laptop but the deployed bot cannot connect at the scheduled time.

How to reduce the risk: Test from the real deployment environment early. Run the worker inside the approved network, use a controlled private connection, or allowlist a stable outbound IP. Include DNS, proxy, TLS certificate, and VPN renewal checks in the readiness test.

Risk that remains: Network maintenance and certificate changes can still cause temporary failures.

6. The automation account may see different data than a human user

HIGH

What it means: Looker and Metabase permissions, row-level security, user attributes, groups, and impersonation rules can change which rows and filters are visible. A service account can accidentally see too much or too little.

What can go wrong: A believable report can contain the wrong department, client, region, or level of detail. Over-permission can also create a data breach.

How to reduce the risk: Create a dedicated group with only the required dashboards and data. Compare automation output with an approved human account for known test cases. Record the service account identity, effective filters, and permission owner.

Risk that remains: Permission changes can silently alter future reports unless regression checks are run.

7. API and export features vary by product, edition, version, and license

HIGH

What it means: Looker provides official asynchronous render tasks and scheduled PNG/PDF delivery [1][2]. Metabase supports scheduled dashboard PDFs and chart alerts, but dashboard subscriptions cannot currently target webhooks [3][4]. Some embedding, filter, branding, and API-key features depend on the edition.

What can go wrong: A design copied from online documentation may not exist in the company's installation or may require a paid license or administrator setting.

How to reduce the risk: Record the exact platform, version, hosting model, license, and enabled administrator settings. Prove one production-like export before designing the rest of the service.

Risk that remains: Upgrades can change behavior; keep source adapters isolated and tested.

8. Metabase API endpoints are not versioned

MEDIUM

What it means: Metabase warns that its API is not versioned and code may need updates when the product changes [5]. An internal endpoint discovered from browser traffic is even more fragile than a documented endpoint.

What can go wrong: An upgrade can break the bot without any change to the bot's own code.

How to reduce the risk: Use documented subscriptions and alerts first. If an API endpoint is required, pin and test the Metabase version, wrap the endpoint in one adapter, and run a compatibility test before upgrades.

Risk that remains: Some maintenance is unavoidable when depending on a changing external product.

C. Rendering and data correctness

9. Dashboard rendering is asynchronous and may take a long time

HIGH

What it means: A report is not ready just because the HTTP page loaded. Charts may be waiting for multiple database queries. Looker's Render API creates a task that must be polled before the result is downloaded [2]. Metabase notes that server-side PDF generation can take a minute or more [3].

What can go wrong: The bot may send blank tiles, loading spinners, partially drawn charts, or time out even though the underlying system is healthy.

How to reduce the risk: Use native render completion states where available. For browser capture, wait for report-specific readiness signals, expected chart counts, and the disappearance of loading/error indicators. Use a clear deadline and fail without sending when the deadline is exceeded.

Risk that remains: Very slow database queries may still miss the delivery window and require report optimization.

10. A successful screenshot can still contain stale data

CRITICAL

What it means: BI systems often cache results. The data pipeline may not have finished, or a dashboard may display yesterday's cached values while looking completely normal.

What can go wrong: The team makes decisions from old data and incorrectly trusts the bot because delivery was on time.

How to reduce the risk: Add a visible or machine-readable data freshness timestamp. Compare it with an agreed cutoff. Trigger the job from a data-ready event when possible, or poll the freshness marker until a deadline. Include the report period in the message caption.

Risk that remains: A fresh timestamp does not guarantee semantic correctness; data-quality checks remain the source owner's responsibility.

11. Fixed delays are unreliable for dynamic dashboards

HIGH

What it means: Sleeping for 10 or 30 seconds assumes every query always completes within that time. Network speed, database load, cache state, and dashboard complexity vary from run to run. Playwright explicitly discourages treating network-idle as a universal readiness test [13].

What can go wrong: Short waits capture incomplete charts; long waits waste time and still fail during exceptional slowness.

How to reduce the risk: Wait for specific DOM states or API responses that prove the required tiles are finished. Validate the screenshot content and stop at a maximum deadline rather than sleeping blindly.

Risk that remains: The dashboard must expose stable readiness signals or the automation will always be partly heuristic.

12. Wrong filters, timezone, locale, or date boundary can produce the wrong report

CRITICAL

What it means: A service account can have different default filters and timezone settings. 'Today' in India is not the same moment as 'today' in UTC. Month-end, daylight-saving changes in other regions, and locale-specific dates can further confuse the result.

What can go wrong: The image looks correct but represents the wrong period, branch, team, or customer.

How to reduce the risk: Store filters and timezone in version-controlled configuration. Use explicit date ranges rather than relative labels where possible. Display the timezone, report period, and important filter values in the output and assert their presence before sending.

Risk that remains: Business definitions such as 'working day' and 'close of day' must still be owned and documented.

13. Dashboard changes can silently break the capture

HIGH

What it means: A title can change, a tile can move, a tab can be renamed, or a designer can replace a component. Browser selectors based on CSS classes or text may then capture the wrong element or nothing at all.

What can go wrong: The bot fails immediately or, worse, sends a cropped or incorrect image without noticing.

How to reduce the risk: Prefer stable dashboard IDs and native render APIs. If browser capture is needed, use stable test IDs or an agreed embed contract. Add a visual smoke test and require dashboard owners to notify the bot owner before structural changes.

Risk that remains: Unannounced layout changes remain a continuing operational risk.

14. Native renderers may not match the interactive dashboard

MEDIUM

What it means: Server-side PDF/PNG generation uses a different rendering environment. Metabase documents differences in theme, map legends, labels, and iframe behavior [3]. Looker documents output-specific layout and row limitations [1]. Fonts, maps, custom visualizations, and embedded content may behave differently.

What can go wrong: The exported report can omit information, move labels, truncate tables, or show unsupported components.

How to reduce the risk: Test every visualization type in the exact export format. Keep a small approved set of dashboards for automation. Replace unsupported tiles or use a controlled browser capture only for those cases.

Risk that remains: Minor visual differences may remain and must be accepted by the report owner.

15. Large tables and multi-tab dashboards do not fit one image

MEDIUM

What it means: A full-page screenshot can become extremely tall. Tables may scroll internally, tabs may not be included, and WhatsApp may scale the image until text is unreadable.

What can go wrong: Important rows or tabs are missing, or recipients must zoom excessively.

How to reduce the risk: Define one phone-oriented report layout. Split the dashboard by section, send a PDF for multi-page content, or send only the most important charts and a secure link to the full report.

Risk that remains: No image format can make a dense analytical application fully usable on a small phone screen.

16. Source database timeouts and load can affect both users and the bot

HIGH

What it means: Rendering a dashboard may run many expensive queries. Scheduling several reports at the same time can create a load spike, exhaust database connections, or hit query timeouts.

What can go wrong: The bot fails and may slow the reporting platform for normal users.

How to reduce the risk: Use caching carefully, stagger schedules, limit concurrency, optimize heavy queries, and coordinate with database owners. Monitor render duration and query failures. Do not retry an expensive report aggressively.

Risk that remains: Peak-load contention can still occur during business-critical periods.

D. Browser automation limitations

17. Saved browser sessions expire and are sensitive secrets

HIGH

What it means: Playwright can reuse authenticated state, but its documentation warns that the state file can contain cookies and headers capable of impersonating the account [10]. Sessions may expire, be revoked, or become invalid after an identity-provider change.

What can go wrong: The bot stops at the login page, or stolen state allows unauthorized dashboard access.

How to reduce the risk: Use API-based authentication when possible. Encrypt any browser state, keep it outside source control and container images, restrict file access, rotate it, and detect login redirects before taking screenshots.

Risk that remains: Browser session renewal may still require a controlled human process.

18. Browser selectors are brittle

HIGH

What it means: CSS classes generated by modern frontends can change between releases. Text selectors break when wording or language changes. Coordinates break when responsive layout changes.

What can go wrong: The automation clicks the wrong tab, applies the wrong filter, or captures an unrelated region.

How to reduce the risk: Use stable roles, labels, test IDs, dashboard IDs, and explicit assertions. Avoid long chains of implementation-specific CSS. Capture diagnostic screenshots and DOM details only on failure, with sensitive information redacted.

Risk that remains: Browser automation remains more maintenance-heavy than an official export API.

19. Headless Chromium can render differently from an employee laptop

MEDIUM

What it means: Fonts, GPU acceleration, screen size, device pixel ratio, locale, and operating-system libraries affect charts and text. A Docker image may not contain company fonts or may render maps differently.

What can go wrong: Text wraps, labels overlap, icons become boxes, or screenshots differ between development and production.

How to reduce the risk: Pin the browser and container versions, install approved fonts, set viewport/locale/timezone explicitly, and visually test output from the real runtime. Use Playwright's locator screenshot capability for stable regions [11].

Risk that remains: Provider updates can still change pixels even with a pinned browser.

20. Chromium consumes significant memory and shared memory

MEDIUM

What it means: Headless browsers are heavier than ordinary API calls. Several parallel dashboards can exhaust a small container. Playwright recommends appropriate Docker init, shared-memory, user, sandbox, and version settings [12].

What can go wrong: Random browser crashes, blank screenshots, zombie processes, and missed schedules appear under load.

How to reduce the risk: Limit concurrency, reuse the browser carefully while isolating contexts, set memory/CPU limits based on measurements, use the recommended container settings, and restart failed workers safely.

Risk that remains: Browser-based capture will always cost more resources than native rendering.

21. CAPTCHA or anti-automation behavior can appear later

HIGH

What it means: An identity provider or application may flag repeated headless logins, new IP addresses, unusual timing, or automated interactions. A CAPTCHA can be introduced after the MVP is deployed.

What can go wrong: Unattended execution stops and cannot recover safely.

How to reduce the risk: Do not attempt to bypass security controls. Use documented service authentication, coordinate with the platform owner, and make the bot fail closed with a clear authentication alert.

Risk that remains: If the platform offers no automation-safe authentication, browser capture may be unacceptable for production.

E. WhatsApp platform and messaging limitations

22. Business onboarding and number registration take external approval

HIGH

What it means: Production use requires a WhatsApp Business Account, an approved sender/phone number, business configuration, and compliance with Meta policies. Twilio also requires a WhatsApp-enabled sender and connection to the Meta business account [8].

What can go wrong: The code may be ready while production access is still unavailable.

How to reduce the risk: Start onboarding during the feasibility phase. Identify who owns the Meta business portfolio and phone number. Use the provider sandbox only for development, not as proof that production approval is complete.

Risk that remains: Approval times and policy decisions are controlled by external parties.

23. Scheduled outbound messages require opt-in and approved templates

CRITICAL

What it means: WhatsApp requires explicit opt-in. Business-initiated messages outside the 24-hour customer service window must use a pre-approved template [8]. A daily report cannot depend on every recipient messaging the bot once per day.

What can go wrong: Free-form image messages may be rejected, recipient complaints may harm account quality, or the account may be restricted.

How to reduce the risk: Document opt-in, opt-out, purpose, frequency, and recipient ownership. Submit an appropriate utility media template with an image or document header. Test the approved production template with real media.

Risk that remains: Meta can reclassify or reject templates, and recipients can block or opt out.

24. The official Groups API has restrictive eligibility and size limits

CRITICAL

What it means: Current provider documentation for Meta's Groups API requires an Official Business Account and at least the 100,000 messaging limit, caps groups at eight participants, and requires invite-based membership [7]. It is unavailable to many small internal teams.

What can go wrong: The exact 'post to our WhatsApp group' requirement may be impossible through the official production API.

How to reduce the risk: Perform an eligibility check before implementation. If the organization is not eligible, recommend individual official delivery or a collaboration tool with supported channel posting.

Risk that remains: Even an eligible group may be too small for the team and may not replace its existing group.

25. Unofficial WhatsApp Web libraries create policy and continuity risk

CRITICAL

What it means: Libraries such as whatsapp-web.js imitate WhatsApp Web. They can reach ordinary groups, but they rely on a logged-in device session and behavior that is not the supported business messaging contract.

What can go wrong: QR sessions log out, protocol changes break the library, the number can be challenged or suspended, and there may be no provider support when the bot fails.

How to reduce the risk: Use only for a time-boxed prototype with a separate test number, no sensitive data, and written risk acceptance. Do not make a critical operational report depend on it.

Risk that remains: No engineering control can turn an unsupported route into a fully supported production service.

26. Media type, size, MIME headers, and hosting rules can reject the report

HIGH

What it means: WhatsApp media messages support specific formats and limits. Twilio documents a 5 MB limit for images and requires the served Content-Type to match the actual file [9]. Template media types are fixed by the approved template header.

What can go wrong: A large PNG, wrong content type, inaccessible URL, or mismatched template causes delivery failure.

How to reduce the risk: Validate dimensions, bytes, MIME type, and extension before sending. Compress or split large images. Prefer provider media upload IDs or short-lived signed URLs reachable by the provider. Do not expose permanent public URLs.

Risk that remains: Provider-side processing may still fail for rare files; keep a PDF or text fallback.

27. Rate limits, account quality, and pricing affect fan-out

MEDIUM

What it means: Sending one report to 50 people creates 50 outbound deliveries. Provider charges, messaging limits, template category, recipient country, and account quality can affect cost and throughput.

What can go wrong: A small pilot becomes expensive or throttled as reports and recipients multiply.

How to reduce the risk: Calculate reports x recipients x frequency before launch. Add budgets and delivery limits. Remove inactive recipients, avoid unnecessary screenshots, and monitor blocks, opt-outs, and quality indicators.

Risk that remains: External pricing and limit models can change.

28. API acceptance is not the same as delivery or reading

HIGH

What it means: The send request can succeed while the message later fails because the recipient is unreachable, opted out, blocked the sender, or the media could not be fetched. Delivered does not always mean read or understood.

What can go wrong: The job is marked successful even though some team members did not receive the report.

How to reduce the risk: Consume status webhooks and keep states such as queued, sent, delivered, failed, and skipped. Define the service objective precisely: accepted, delivered, or read. Alert on terminal failure and provide a safe manual resend.

Risk that remains: Read receipts may be unavailable or disabled, and human understanding cannot be guaranteed.

29. The sending number and business account need clear ownership

HIGH

What it means: A personal employee number is not a durable production asset. Employees leave, phones are replaced, ownership disputes occur, and credentials may be tied to one administrator.

What can go wrong: The bot becomes unmanageable or loses access when the original creator is absent.

How to reduce the risk: Use a company-owned number and business portfolio with at least two approved administrators. Document number recovery, provider access, billing ownership, and emergency disablement.

Risk that remains: External account recovery can still take time, so a fallback channel is necessary.

F. Security, privacy, and governance

30. Screenshots can leak confidential information more easily than dashboards

CRITICAL

What it means: A dashboard normally applies login and permission checks each time it is opened. A screenshot is a detached copy. It can be forwarded, downloaded, backed up, photographed, or remain on a former employee's device.

What can go wrong: Customer data, revenue, employee information, system incidents, or commercial metrics escape the controlled BI environment.

How to reduce the risk: Classify every report. Remove unnecessary PII and sensitive columns, restrict recipients, watermark with date/classification, obtain data-owner approval, and prefer a secure link when screenshots are not allowed. Define an incident process.

Risk that remains: Once delivered to a personal device, technical control over forwarding is limited.

31. Public report links or media URLs can bypass access control

CRITICAL

What it means: Metabase public links have no authentication and anyone with the URL can view the data [6]. Media URLs used by a messaging provider may also be exposed if they are permanent or guessable.

What can go wrong: A copied link gives unauthorized people access even after recipient removal.

How to reduce the risk: Do not use public dashboard links for sensitive content. Use authenticated or signed embeds, provider media uploads, or short-lived signed storage URLs with narrow permissions. Log access without logging the URL token.

Risk that remains: A recipient can still copy the downloaded media itself.

32. Secrets can leak through source code, container images, logs, and screenshots

CRITICAL

What it means: BI client secrets, API keys, WhatsApp tokens, session cookies, signed media URLs, phone numbers, and report contents are all sensitive. Debug logging can expose them even if code does not.

What can go wrong: Attackers can impersonate the bot, read reports, send messages, or create unexpected charges.

How to reduce the risk: Use a managed secret store, least-privilege tokens, rotation, masked structured logging, repository secret scanning, restricted runtime identities, and separate test/production credentials. Never bake secrets or browser state into the image.

Risk that remains: Privileged operators still require governance and audit.

33. Temporary files and logs create a hidden data-retention system

HIGH

What it means: Screenshots may be written to local disk, object storage, message-provider storage, backups, failed-job folders, and observability tools. Each copy can outlive the business need.

What can go wrong: Sensitive reports remain accessible long after delivery and increase breach scope.

How to reduce the risk: Map every storage location and provider retention rule. Encrypt in transit and at rest, apply automatic expiry, delete temporary files after use, and keep only minimal audit metadata such as checksum, report ID, time, and delivery state.

Risk that remains: Provider and device backups may have retention outside the bot's direct control.

34. Recipient membership changes over time

HIGH

What it means: Phone numbers stored in configuration become stale. People change roles, leave the company, share devices, or no longer need the report.

What can go wrong: Former or unauthorized recipients continue receiving confidential information.

How to reduce the risk: Use an owned access list with an approval trail, periodic review, immediate removal workflow, and a clear owner. Do not hard-code phone numbers in source. Reconcile the list before sensitive sends.

Risk that remains: Role-change signals must arrive promptly from the identity or HR process.

35. Legal and policy requirements vary by organization and country

HIGH

What it means: Consent, employee monitoring, cross-border transfer, retention, financial controls, healthcare rules, and customer contracts may apply depending on the report. A technically possible design is not automatically permitted.

What can go wrong: The project creates compliance, contractual, or disciplinary exposure.

How to reduce the risk: Have security, privacy, legal, and data owners review the specific report and recipient model. Document lawful purpose, consent/notice, processor/provider roles, location, retention, and incident handling. This report is technical guidance, not legal advice.

Risk that remains: Policy approval must be revisited when report content, countries, or providers change.

G. Scheduling and reliability

36. In-process cron can miss jobs or run them twice

HIGH

What it means: A library such as node-cron only runs while that process is alive. Deployments, crashes, autoscaling, or two replicas can skip a job or execute it twice.

What can go wrong: The report is absent before stand-up or appears multiple times.

How to reduce the risk: Use a managed scheduler to create a durable job, a database/queue for state, and a distributed lock. Give each job an idempotency key such as report ID + business date + recipient.

Risk that remains: The scheduler itself can fail, so late/missing-job monitoring is still required.

37. Clock time is not the same as data readiness

CRITICAL

What it means: A 9:50 AM schedule assumes that upstream data is complete before 9:50 AM every day. Weekends, month-end processing, late files, and failed ETL runs break that assumption.

What can go wrong: The bot sends incomplete data exactly on time.

How to reduce the risk: Prefer an event from the data pipeline. Otherwise check a freshness table/metric and wait with a deadline. If data is not ready, send no report and alert the owner with the exact reason.

Risk that remains: The business must choose between timeliness and completeness when upstream processing is late.

38. Timezone, daylight-saving, holidays, and business-day rules cause schedule errors

MEDIUM

What it means: Servers often run in UTC while the team works in Asia/Kolkata or another region. Some recipients may observe daylight-saving time. Public holidays and month-end schedules differ.

What can go wrong: Reports arrive an hour early/late, on non-working days, or for the wrong business date.

How to reduce the risk: Store an IANA timezone, not just a numeric offset. Define business calendar behavior, show the calculated next run, and test month/year boundaries. Keep report timezone separate from delivery timezone when needed.

Risk that remains: Business calendars require maintenance by an owner.

39. Retries can create duplicate messages

HIGH

What it means: A network timeout does not prove the provider rejected the request. Retrying may submit the same message again. Multiple workers can also process the same job.

What can go wrong: Recipients receive duplicate reports and lose confidence in the bot.

How to reduce the risk: Persist an idempotency key and provider message ID before acknowledging the job. Retry only known transient failures with exponential backoff and a limit. Check existing state before every resend.

Risk that remains: Exactly-once delivery across external systems is difficult; the practical goal is idempotent intent and duplicate detection.

40. A fan-out job can partially succeed

HIGH

What it means: When sending individually, nine people may receive the report while one fails. Treating the entire batch as one success/failure loses detail.

What can go wrong: Some recipients are silently missed, or successful recipients get duplicates during a full-batch retry.

How to reduce the risk: Create one delivery record per recipient under one report job. Retry only failed recipients. Present a summary such as 9 delivered, 1 failed, with the failed identity visible only to authorized operators.

Risk that remains: The team needs a policy for whether partial delivery is acceptable before a meeting.

41. Provider, network, storage, or BI outages can overlap

HIGH

What it means: The system depends on several services. A render may succeed while storage fails; storage may succeed while WhatsApp is unavailable; delivery may work while the status webhook is delayed.

What can go wrong: Failures become difficult to diagnose and can leave orphaned sensitive files.

How to reduce the risk: Give every stage its own status and timeout. Use bounded retries, a dead-letter/manual-review state, cleanup jobs, and a fallback notification channel. Do not hide an upstream error behind a generic 'bot failed' message.

Risk that remains: A multi-provider pipeline cannot eliminate correlated outages.

42. Without observability, the team learns about failure during the meeting

HIGH

What it means: A silent background task needs evidence of what happened: schedule created, render started, validation passed, send accepted, delivered, failed, or cleaned up.

What can go wrong: The report is missing and nobody knows whether the problem is data, login, rendering, WhatsApp, or scheduling.

How to reduce the risk: Use structured logs, metrics, stage timings, correlation/job IDs, status dashboards, and actionable alerts. Send alerts to a channel independent of WhatsApp. Define business and technical owners plus a manual resend runbook.

Risk that remains: Alerts still require a person who is available and empowered to act.

H. User experience and communication

43. Phone screens and WhatsApp compression make dense dashboards unreadable

HIGH

What it means: A desktop dashboard designed for a wide monitor shrinks dramatically on a phone. Image compression, long tables, small legends, and pale colors make the problem worse.

What can go wrong: Users zoom repeatedly, misread numbers, or ignore the report.

How to reduce the risk: Create a dedicated mobile report layout with large type and fewer KPIs. Use a controlled viewport, split sections, test on common phones, and keep images within provider limits [9].

Risk that remains: Some analytical detail will still require the original dashboard or a PDF.

44. An image without context can be misunderstood

HIGH

What it means: Recipients may not know the report period, timezone, filters, units, whether data is preliminary, or why a metric changed.

What can go wrong: The team discusses the wrong period or treats preliminary numbers as final.

How to reduce the risk: Include a concise caption with report name, data period, generated time, timezone, filters, freshness status, classification, and a secure source link. Add a warning when data is delayed or partial rather than sending a normal-looking image.

Risk that remains: A caption cannot replace training and metric definitions.

45. Screenshots are poorly accessible

MEDIUM

What it means: Screen readers cannot interpret chart text inside an image. Color-only status indicators may not work for color-blind users, and small text is difficult for low-vision users.

What can go wrong: Some team members cannot independently use the report.

How to reduce the risk: Send a short text summary of key metrics and status alongside the image. Use clear labels, high contrast, and accessible source reports. Provide a PDF/text alternative when required.

Risk that remains: Charts still need proper alternative explanations outside the screenshot.

46. Too many automated reports create notification fatigue

MEDIUM

What it means: If every dashboard sends several images every day, users stop opening them. Repeated non-actionable information lowers trust and encourages muting the business sender.

What can go wrong: Important alerts are overlooked and account quality can decline through blocks or opt-outs.

How to reduce the risk: Define the decision each report supports, combine related charts, send only on business days or meaningful change where appropriate, and review usage with recipients after the pilot.

Risk that remains: Information needs change, so periodic pruning is required.

I. Operations, maintenance, and cost

47. External libraries and APIs change over time

HIGH

What it means: Looker, Metabase, Playwright, browser images, WhatsApp API versions, Twilio, and operating-system packages all evolve. An automatic latest-version upgrade can break compatibility; never upgrading creates security risk.

What can go wrong: A routine dependency update breaks rendering or sending, or an old dependency becomes unsupported.

How to reduce the risk: Pin versions, use supported releases, schedule controlled upgrades, run integration tests against a staging dashboard and sandbox recipient, and monitor vendor deprecation notices.

Risk that remains: Maintenance effort is permanent, not a one-time project cost.

48. Costs multiply across rendering, storage, messaging, and support

MEDIUM

What it means: Costs include BI licenses/service accounts, database queries, browser compute, object storage/egress, provider fees, WhatsApp messages per recipient, monitoring, and engineering support.

What can go wrong: A seemingly small automation becomes expensive as dashboards, recipients, and frequency increase.

How to reduce the risk: Create a cost model for pilot and expected scale. Track cost per report run, expire artifacts, cap recipients and frequency, and compare with native email/Slack delivery.

Risk that remains: Pricing can change and support effort is difficult to predict exactly.

49. Testing only the happy path gives false confidence

HIGH

What it means: A demo usually uses a valid session, warm cache, complete data, one recipient, and a healthy network. Production failures occur in expired sessions, partial rendering, wrong filters, large media, duplicate jobs, and provider errors.

What can go wrong: The service fails immediately when a realistic exception occurs.

How to reduce the risk: Automate unit, contract, integration, visual, security, and failure-path tests. Simulate expired credentials, login redirects, slow tiles, stale data, 429/5xx responses, storage failure, duplicate scheduler events, and partial delivery.

Risk that remains: External sandbox behavior may not perfectly match production.

50. No operating owner means the bot will decay

CRITICAL

What it means: Someone must own report definitions, recipients, credentials, vendor accounts, budgets, alerts, upgrades, and incident decisions. A project handed over without these roles becomes orphaned.

What can go wrong: Failures persist, access remains over-broad, and nobody is authorized to rotate credentials or change recipients.

How to reduce the risk: Name a business owner, report/data owner, platform owner, security approver, and technical operator. Document access, dashboards, schedules, escalation, manual resend, disablement, recovery, and review dates.

Risk that remains: Ownership must be updated during reorganizations and staff changes.

6. Platform-specific implementation approach

6.1 Looker

Looker is the strongest case for avoiding browser automation. It supports scheduled dashboard delivery to built-in destinations such as email, webhook, S3, and SFTP, with PNG/PDF formats depending on the destination [1]. Its API can create a dashboard render task, poll the task, and download PNG, JPG, or PDF output [2].

- Use a dedicated Looker API/service identity with only the required models, folders, and dashboards.
- Pass explicit dashboard filters and layout settings; do not rely on a human user's defaults.
- Poll the render task with bounded backoff and an overall deadline.
- Validate the downloaded MIME type, dimensions, file size, dashboard identity, and freshness marker.
- Test maps, long tables, custom visualizations, and multi-tab dashboards in the chosen format.

6.2 Metabase

Metabase has several useful native paths, but they differ by report type. A question alert can send a webhook containing a base64 PNG visualization [4]. A full dashboard subscription can send email or Slack and attach a server-generated PDF [3], but dashboard subscriptions cannot currently send to a webhook [4].

- For one important chart, prefer an alert webhook and decode the supplied PNG.
- For a full dashboard where PDF is acceptable, send the subscription to a controlled bot mailbox and process the attachment.
- If a PNG is mandatory, convert approved PDF pages to images or capture a secure authenticated view with Playwright.
- Do not expose sensitive dashboards with a public link; public links do not provide authentication [6].
- Treat undocumented API/browser endpoints as version-sensitive and test before every Metabase upgrade [5].

6.3 Playwright fallback

Use Playwright only when the platform cannot provide an acceptable native artifact. It supports full-page and element screenshots [11], but its saved authentication state is sensitive [10] and its container/runtime needs careful configuration [12].

- Set viewport, device scale, timezone, locale, fonts, and theme explicitly.
- Use stable locators and assert the expected dashboard title, filters, and tile count.
- Wait for explicit readiness; do not use only a fixed delay or network-idle [13].
- Capture diagnostics only on failure and protect them like production reports.
- Run browser work with limited concurrency and pinned browser/container versions.

6.4 WhatsApp delivery

The default production design should send one approved media template per opted-in recipient through Meta Cloud API or a provider such as Twilio. Twilio documents opt-in, business-initiated template use, and the 24-hour customer service window [8]. Media must meet supported format, MIME, and size rules [9].

- Store recipient consent, purpose, frequency, locale, and opt-out state.
- Approve an image-header or document-header utility template before launch.
- Upload media through the provider or use a short-lived signed URL; never a permanent public file.
- Create a delivery record per recipient and consume status webhooks.
- Use the official Groups API only after eligibility and group design are confirmed [7].

7. Failure scenarios and expected response

Scenario	How to detect it	Safe response
Login redirects to SSO	Expected dashboard title missing; login URL detected	Stop. Alert authentication owner. Do not capture or send.
One or more tiles still loading	Spinner/error markers or expected tile count missing	Wait to deadline. Mark render failed. Do not send partial report.
Data timestamp is stale	Freshness value is older than cutoff	Skip delivery, alert data owner, optionally retry until business deadline.
Media is over limit	Byte/MIME/dimension validation fails	Split or compress deterministically; otherwise send approved PDF/text fallback.
WhatsApp returns rate limit	HTTP 429/provider code	Bounded backoff, preserve job state, avoid immediate fan-out retry.
Send call times out	No definitive provider response	Query existing provider/job state before retry to avoid duplicates.
Some recipients fail	Per-recipient status webhook	Retry only failed recipients; alert with authorized summary.
Provider outage	Health/status errors across jobs	Pause retries, use fallback channel, communicate expected delay.
Dashboard structure changes	Visual/DOM contract test fails	Disable automated send and request dashboard owner review.
Cleanup fails	Artifact remains after TTL	Run cleanup alert/job; block public exposure; investigate storage permissions.

Fail-closed rule

If the service cannot prove the report is correct, current, complete, and authorized, it should alert and skip delivery. It should never send a questionable artifact merely to meet the clock.

8. Phased implementation plan

1	Phase 0 - Requirements and governance Confirm destination, platform/version/license, exact report, filters, timezone, freshness rule, network path, data classification, recipients, opt-in, owners, retention, and fallback.
2	Phase 1 - Two-boundary proof From the actual production-like runtime, render one approved report and send one approved test media template. This proves the two hardest external boundaries before building infrastructure.
3	Phase 2 - Minimal viable service Implement source adapter, artifact validator, image/PDF preparation, delivery adapter, durable job ledger, scheduler entry point, status webhook, secrets, and alerts.
4	Phase 3 - Hardening Add idempotency, bounded retries, per-recipient states, freshness gate, visual contracts, security controls, artifact TTL, dashboards, runbooks, and provider failure handling.
5	Phase 4 - Pilot Run at least 30 scheduled production-like jobs to test real timing, data readiness, phone readability, partial delivery, ownership, and support response. Keep recipients limited.
6	Phase 5 - Production Complete security/privacy approval, enable production recipients, monitor success and costs, review recipient access, rehearse recovery, and schedule dependency/platform upgrades.

9. Testing strategy

Test area	Evidence required
Requirements	Approved sample output, exact recipients, schedule, filters, timezone, freshness rule, and fallback.
Functional	Looker/Metabase render, image/PDF preparation, template send, webhook status, manual resend.
Correctness	Known-date and known-filter fixtures; compare output with approved human report.
Failure path	Expired credentials, login page, stale data, spinner, query error, oversized media, 429/5xx, timeout.
Idempotency	Duplicate scheduler event, worker restart, webhook replay, ambiguous send timeout.
Security	Least privilege, secret scan, log redaction, signed URL expiry, recipient removal, artifact deletion.
Visual	Desktop and phone readability, fonts, maps, tables, multiple tabs, compression.
Load	Expected reports x recipients, staggered schedules, database impact, browser memory, provider throughput.
Operations	Alerts reach independent channel; runbook, rotation, disablement, recovery, and ownership drill.
Pilot	Thirty consecutive scheduled runs accounted for with no unexplained duplicate or silent failure.

10. Production acceptance criteria

- The artifact contains the expected report title, period, timezone, filter values, and all required tiles.
- The data freshness marker is newer than the approved cutoff, or the job is skipped with an alert.
- No report is sent from a login page, error page, loading state, blank image, or stale cache.
- The service account has only the access needed for the approved report.
- Every recipient has documented authorization and opt-in; removal can be completed quickly.
- The WhatsApp media template, file format, MIME type, size, and real production sender have been tested.
- Every scheduled job and recipient delivery has an auditable state and job ID.
- Duplicate schedule events, restarts, timeouts, and webhook replays do not create duplicate sends.
- Artifacts are encrypted and deleted within the approved retention period.
- Alerts, manual resend, credential rotation, provider outage, disablement, and recipient removal runbooks are tested.
- Business, report/data, security, platform, and technical operations owners are named.
- Thirty consecutive pilot runs are accounted for with no silent failure or unexplained duplicate.

11. Questions to ask the manager and platform owners

- Must the destination be an existing WhatsApp group, or are individual messages acceptable?
- If a group is mandatory, is the company an Official Business Account and eligible for the current Groups API?
- Which BI product, hosting model, version, and license are in use?
- Which exact dashboard, tab, charts, filters, timezone, and data period must be delivered?
- Is PDF acceptable, or is a PNG screenshot mandatory?
- How do we know the upstream data is complete, and what happens when it is late?

- Does access require SSO, MFA, VPN, trusted device, private DNS, or IP allowlisting?
- What data classification applies? Does the report contain PII, financial, customer, employee, or security information?
- Who approves and owns the recipient list, opt-in, opt-out, and offboarding?
- How long may screenshots and logs be retained, and in which countries/providers may they be processed?
- What delivery success is required: API accepted, delivered, read, or acknowledged?
- What is the acceptable delay, retry window, and fallback channel?
- Who pays for and administers the BI account, WhatsApp business account, phone number, messaging provider, and cloud runtime?
- Who responds when the report is missing before stand-up?

12. Final recommendation

Recommended production path

Use TypeScript/Node.js with a native Looker or Metabase source adapter, an artifact validator, phone-oriented image/PDF preparation, a managed scheduler plus durable job ledger, official one-to-one WhatsApp Business delivery, per-recipient status webhooks, independent alerts, and short-lived encrypted storage.

Do not begin by automating WhatsApp Web. Begin by confirming the destination and proving the source export. If the company insists on an existing normal group but does not qualify for the official Groups API, report that as a platform limitation and recommend individual delivery or Slack/email. That is a legitimate research conclusion, not a failure of the development team.

For Looker, use the official render/scheduling capabilities first [1][2]. For Metabase, use alert webhooks for individual chart PNGs [4], subscriptions for full-dashboard PDFs [3], and Playwright only when an exact whole-dashboard PNG is essential. Regardless of source, a report should be sent only after freshness, filters, completeness, format, authorization, and recipient checks pass.

13. Glossary

Term	Simple meaning
API	A supported way for one program to request data or actions from another program.
Artifact	The PNG, JPG, or PDF report file created for delivery.
BI tool	Business intelligence software such as Looker or Metabase that displays data as dashboards.
Customer service window	The 24-hour WhatsApp period after a user messages a business, during which certain free-form replies are allowed.
Fail closed	Do not send when correctness or authorization cannot be proven.
Idempotency	Repeating the same job request does not produce an unintended duplicate message.
MIME type	A label such as image/png that tells systems what kind of file is being transferred.
OBA	Official Business Account on WhatsApp; a higher-trust business status needed for current Groups API eligibility.
Opt-in	Clear permission from a recipient to receive WhatsApp messages from the business.
Playwright	A browser automation tool that can log in, navigate pages, and take screenshots.
Render	Generate a final visual form of a dashboard as an image or PDF.
Service account	A non-human account created for automation with limited permissions.
Signed URL	A temporary link containing authorization that expires after a short time.
Status webhook	An HTTP notification sent by the provider when a message changes state, for example delivered or failed.
Template	A WhatsApp message format submitted for approval and used for business-initiated messages.

14. References

Sources were checked for the current platform behavior described in this report. Product rules and APIs can change; re-check them during implementation and before each major upgrade.

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https://docs.cloud.google.com/looker/docs/reference/looker-api/latest/methods/RenderTask/create_dashboard_render_task
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- [13] **Playwright - waitForLoadState and network-idle warning**
<https://playwright.dev/docs/api/class-frame#frame-wait-for-load-state>

Scope note: The WhatsApp screenshots and pasted assignment supplied by the user were treated as project context. Their text was not treated as an instruction to the report author.

Disclaimer: This report provides technical and operational guidance. It is not legal advice. Organization-specific security, privacy, employment, contractual, and regulatory decisions require the appropriate internal reviewers.